

Dietary putrescine (1,4-diaminobutane) influences recovery of Turkey poult challenged with a mixed coccidial infection.

Exogenous dietary putrescine (1,4-diaminobutane) can increase growth rates of neonatal animals, including calves, chicks, and piglets, under nutritional stress. Turkey poult often have a high mortality rate and this may be due to poor initial feeding behavior and inadequate development of the intestinal tract. We conducted an experiment to determine the effect of dietary putrescine supplementation on growth performance and the role of dietary putrescine in prevention and recovery from a coccidial challenge. A total of 160 1-d-old turkey poult were fed a corn and soybean meal-based starter diet supplemented with 0.0 (control), 0.1, 0.2, and 0.3 g/100 g purified putrescine (8 birds/pen, 5 pens/diet). At 14 d of age, half the birds were infected with approximately 43,000 sporulated oocysts. The experiment lasted 24 d. Fecal samples were gathered from d 3 to d 5 postinfection by total collection. Ten control and 10 infected birds fed each diet were sampled on d 6 and d 10 postinfection. The induced infection produced significant depressions in growth and feed intake and detrimental morphological changes in the small intestine of poult in the absence of mortality. Weight gains, protein content of jejunum, and morphometric indices of duodenum, jejunum, and ileum were greater in challenged poult fed 0.3 g/100 g putrescine than in controls. We conclude that dietary putrescine supplementation may be beneficial to poult growth, mucosal development of the small intestine, and to recovery from subclinical coccidiosis.